

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
SEVERE SERVICE VALVE DIAGNOSTIC INSPECTION - VDU VACUUM SECTION

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Equipment ID: MRPL-VALVE-402-MOV

Equipment Description: Vacuum Tower Bottoms Emergency Isolation & Letdown Valve

Operating Unit: VDU-1 Vacuum Residue Transfer System

Inspection Date: 2026-09-07

Inspector: D. Fernandez (Control Valves & Actuation Reliability Team)

FIELD DIAGNOSTIC OBSERVATIONS & SIGNATURE ANALYSIS:

- Actuator Motor Running Torque Margin: Dropped to 14% (Minimum required safety margin: 30%).
- Valve Full Stroke Time (Closed to Open): 48.5 seconds (Allowable limit: <= 25.0 seconds).
- Acoustic Emission Sensor (AE): Ultrasonic noise 84 dB at 38 kHz (Severe cavitation erosion).
- Stem Packing Gland Zone: Severe graphite packing extrusion with active pitch tar weeping.
- Seat Leakage (FCI 70-2 Class VI Test): Passing 4.2 L/min against zero-leakage requirement.
- Internal Trim Condition: Stellite-faced plug and cage exhibit micro-pitting and washouts.

SAFETY & COMPLIANCE EVALUATION: LEVEL 2 HIGH ALERT

Sluggish emergency closure and seat leakage under 380 deg C vacuum residue service risks fire hazard upon downstream transfer line isolation during emergency unit de-pressurization.

ACTION DIRECTIVE:

1. Re-torque actuator bevel gearbox packing bolts immediately to restore temporary seal.
2. Prepare spare tungsten carbide cage trim assembly (Type 410 / Satellite 6).
3. Schedule isolation valve hot-swap during upcoming planned maintenance window.
4. Overhaul Limitorque electric actuator gearbox and replace degraded worm drive bearings.